Najd Rahmani

Final-Year Data Science Engineering Student – Agentic AI / Data Engineering

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Portfolio · Github · LinkedIn

PROFILE SUMMARY

Final-year Data Science Engineering student at ESPRIT building agentic AI and data systems: multi-agent pipelines with validation steps, cloud monitoring on Azure and Microsoft Foundry, and Python/FastAPI services. Seeking a 6-month final-year internship (PFE) in agentic AI and automation.

PROFESSIONAL EXPERIENCE

CED Group *Tunisia*

Intern *Jul 2026 – Sep 2026*

- Designed and implemented a **data pipeline** (**Azure Cost Management & Monitor APIs** → **SQL** → **Flutter dashboard**) that reduced data retrieval time versus direct API querying, with Microsoft authentication and dynamic resource-group/date-range filtering.
- Built and deployed a cloud monitoring and cost-analysis dashboard for **Azure and Microsoft Foundry**, tracking usage, latency, token consumption, and cost.

PROJECTS

Recipe Helper – Multimodal Agentic RAG Assistant *(n8n, Qdrant, Gemma3 vision, LoRA)* *2026*

- Built a multi-agent pipeline that returns verified recipes from photos or text: a **supervisor agent** orchestrates Search/Embed, Rank/Chef and **Validator** tools, and a deterministic formatter releases only validator-approved output.

Delivery Lab – Local Multi-Agent Delivery Platform *(Python, Ollama)* *2026*

- Built a local lab of **31 specialised agents** exposed through a web interface and API, running on fictional data or against a local **Ollama** model.
- Enforced evaluation rigor: chronological train/validation/test splits, **leakage checks** on feature and label timestamps, versioned models, and **11 regression tests**.

ARDHI – Smart Agricultural AI Platform *(team project, ESPRIT)* *2025 – 2026*

- Team-built platform for crop disease detection (EfficientNet-B0), irrigation planning, fertilizer recommendations, and real-time livestock monitoring over **WebSocket**.
- Role: Data Scientist – developed **5 computer vision models** for livestock health and welfare, served through a **FastAPI** microservice following the TDSP methodology.

Breast Cancer ML Diagnostic Platform *2024 – 2025*

- End-to-end ML/DL solution on FNA cellular data: malignant/benign prediction, feature-importance explainability, and Low/Medium/High risk stratification, with role-based dashboards for patients, doctors, and admins.

EDUCATION

Private Higher School of Engineering and Technology (ESPRIT) *Ariana, Tunisia*

Engineering Degree – Data Science Specialization *2022 – 2027 (expected)*

Relevant coursework: Big Data, Data Science, Machine Learning, Deep Learning, Generative AI, Explainable AI.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++
Agentic AI & LLMs: Multi-agent orchestration, RAG, LLMs, Ollama, Generative AI, n8n
Data Engineering: Data Pipelines (ETL/ELT), REST API Ingestion, PostgreSQL, Vector Databases (Qdrant)
Machine Learning: Deep Learning, Computer Vision, Scikit-learn, PyTorch, TensorFlow, MLflow
Cloud & BI: Microsoft Azure (Cost Management, Monitor, Foundry), AWS, Power BI
Backend & Tools: FastAPI, Flask, Docker, Git, Flutter, regression testing

CERTIFICATIONS

AWS – Certified Cloud Practitioner · **NVIDIA** – Fundamentals of Deep Learning, AI Applications for Anomaly Detection, Transformer-Based NLP Applications · **Hedera** – HBF, HCF, HCDA

LANGUAGES

Arabic: Native · **French:** Professional Proficiency · **English:** Professional Proficiency

LEADERSHIP & ACTIVITIES

JTI – Junior Chamber International Tunisia: Tunis chapter member – leadership development and community initiatives.
Esprit Google Club & Esprit Data Science Club: Member – AI/ML workshops, knowledge-sharing, and collaborative projects.